ANAT20060
Locomotor Biology

TOPICS

Introduction to Module

Fascia of the lower limb

Text

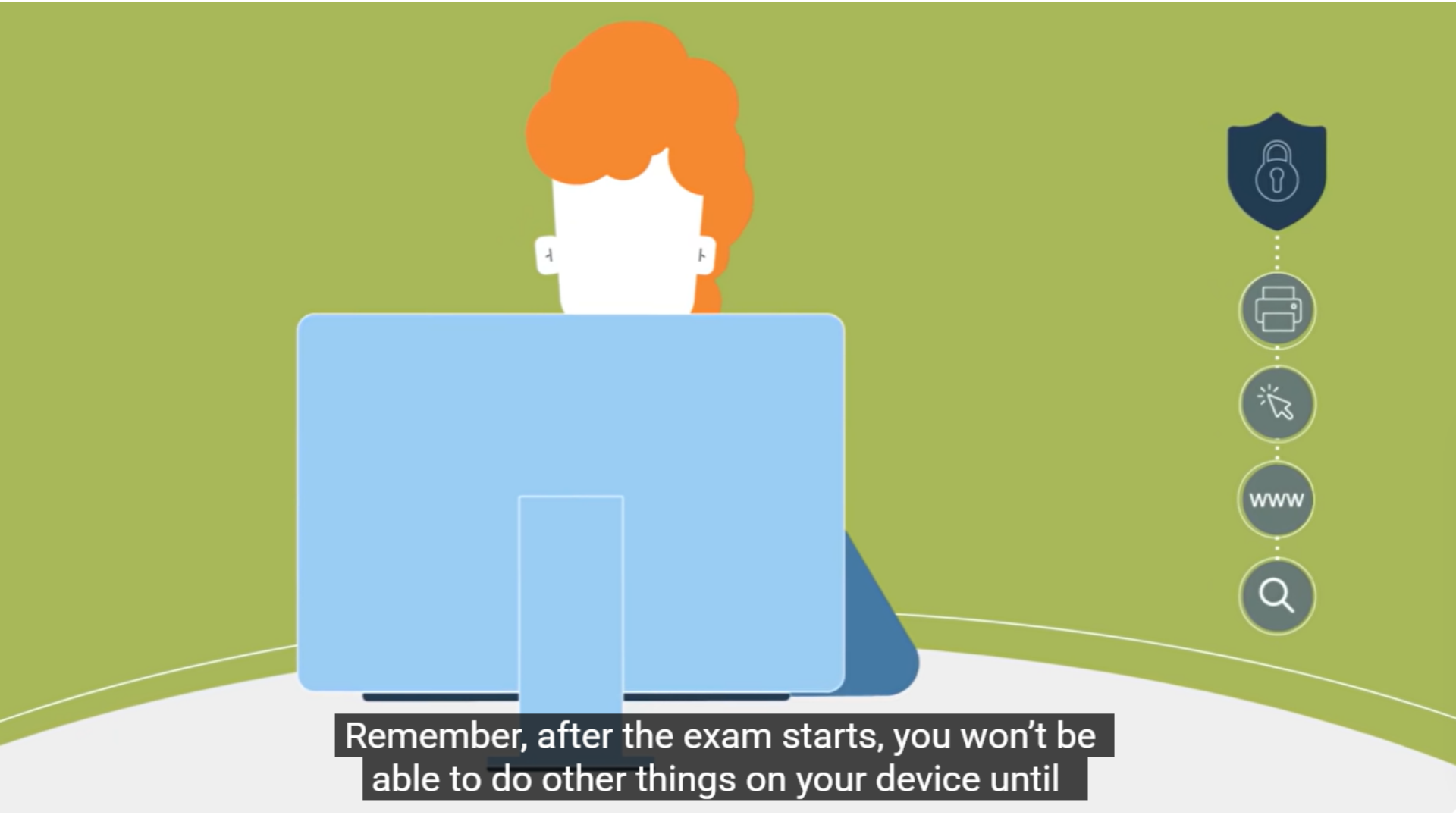
Module assessment

Assessment Strategy for ANAT20060

All modules require at least one component and all fields within that component are mandatory.

Assessment Type	Description	Text Schedule Details	% of Final Grade	Component Scale
Exam (Online)	Ten short answer questions on Brightspace. Inivigiliated in Lecture theatre setting.	Week 8	20.00	Alternative linear conversion grade scale 40%
Exam (In-person)	50 question MCQ (USMLE style)	Location:Exam Centre Duration:1 hr(s)	80.00	Alternative linear conversion grade scale 40%
Total			**100.00**	

“Respondus” Lock Down Browser

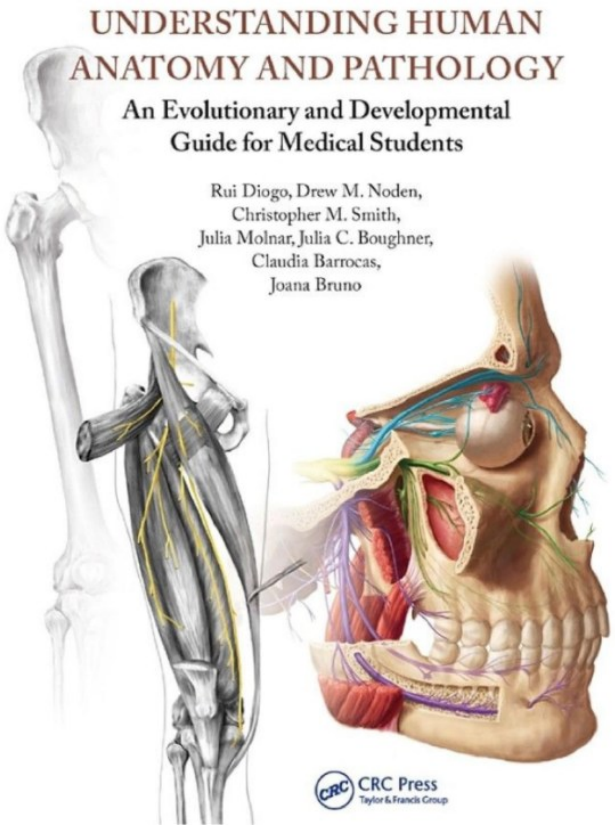


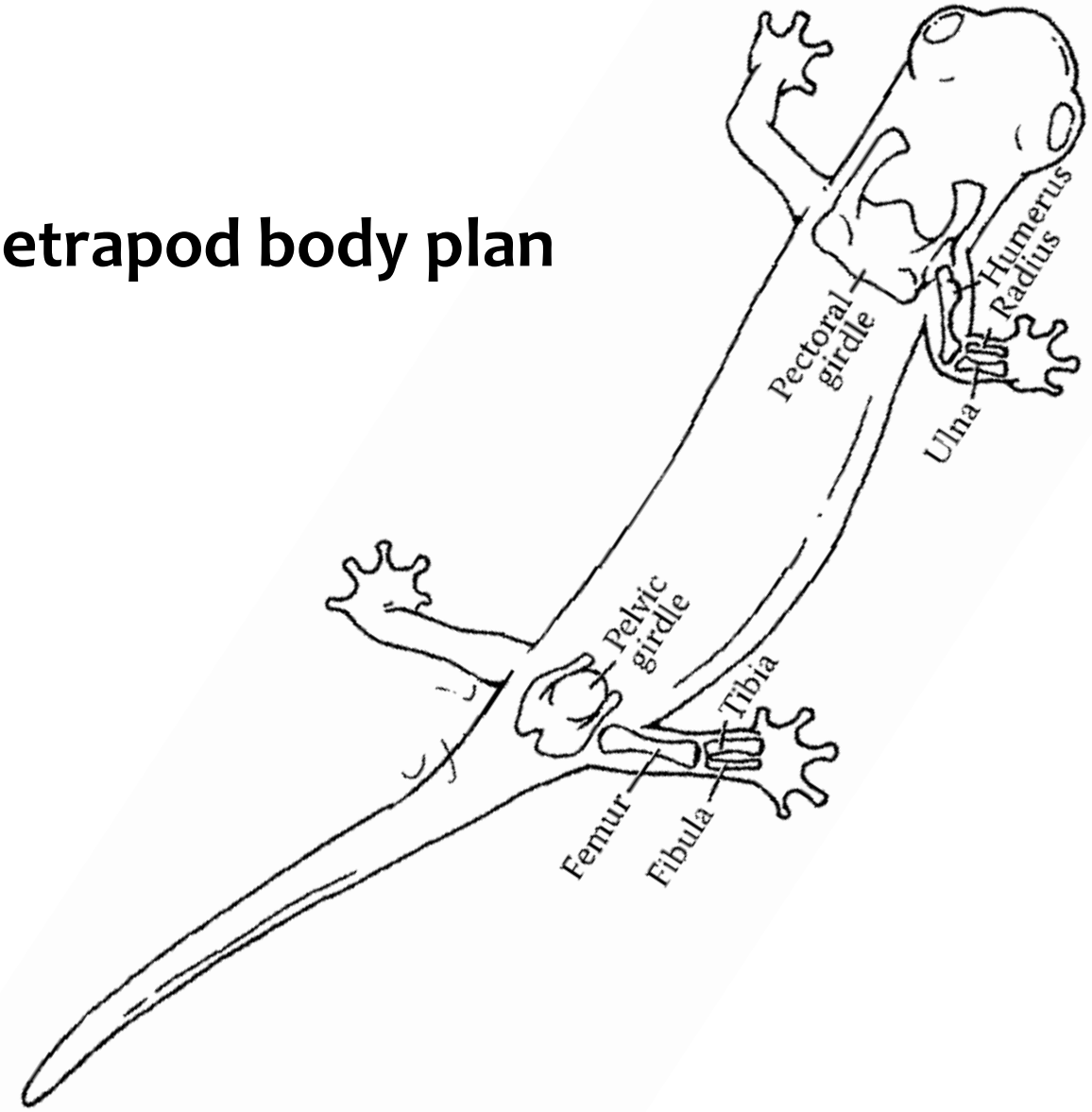
https://web.respondus.com/lockdownbrowser-student-video/

Learning Objectives of Lecture

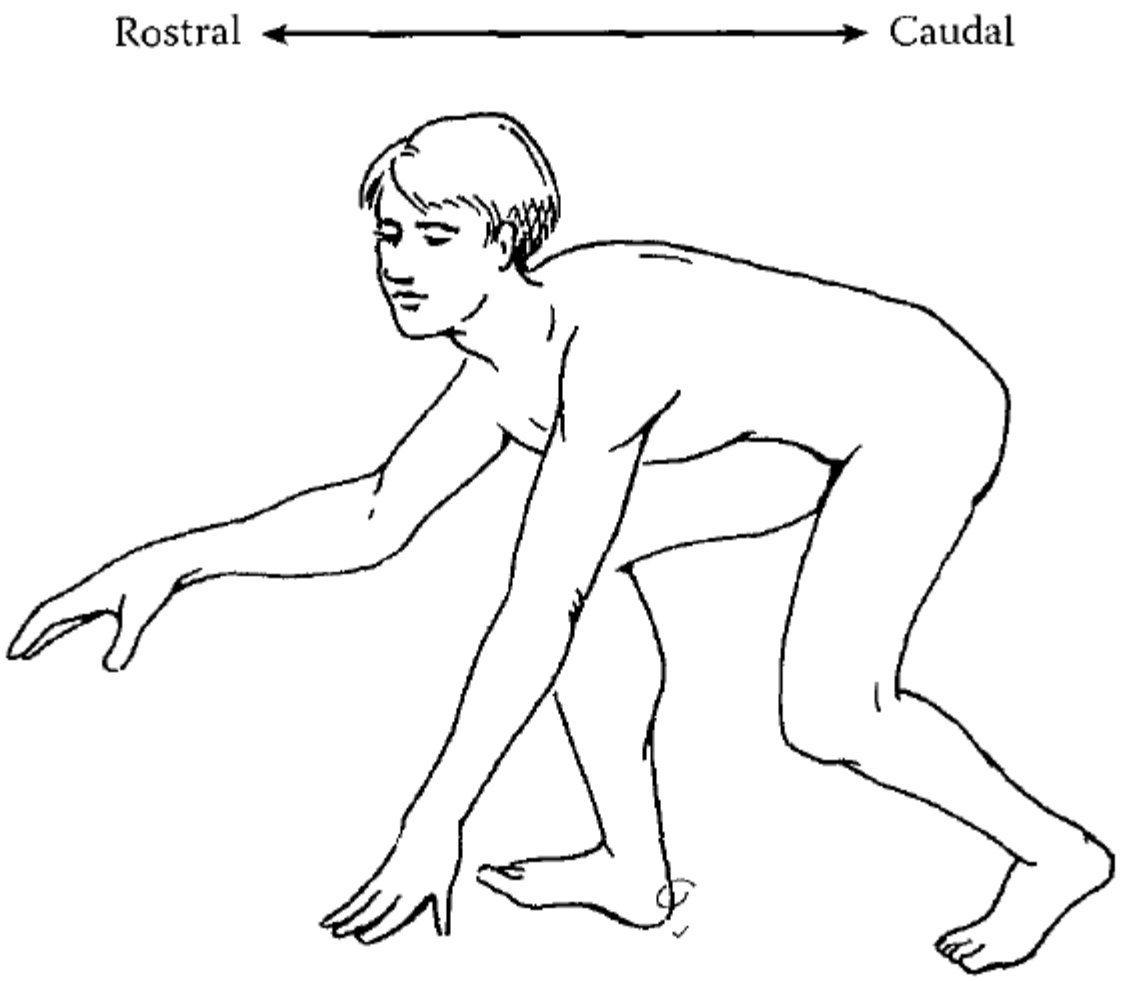
1. Appreciate the similarities and differences between upper and lower limbs
2. Describe the great fasciae of the lower limb: Gluteal fascia, Fascia lata, Crural fascia
   butt thighs 'leg' = under knee
3. Understand how abdominal fascia blends with lower limb fascia
4. Apply anatomical knowledge to diagnose and localise urethral injury
5. Understand how the muscle “Tensor Fasciae Latae” operates with other hip abductors
6. Be able to perform and interpret a Trendelenberg test.
7. Enumerate the fascial compartments of thigh and leg
8. Apply this knowledge to “compartment syndrome”
9. Describe the lesser fasciae of lower limb : cribriform, popliteal, plantar.
   smaller

The tetrapod body plan

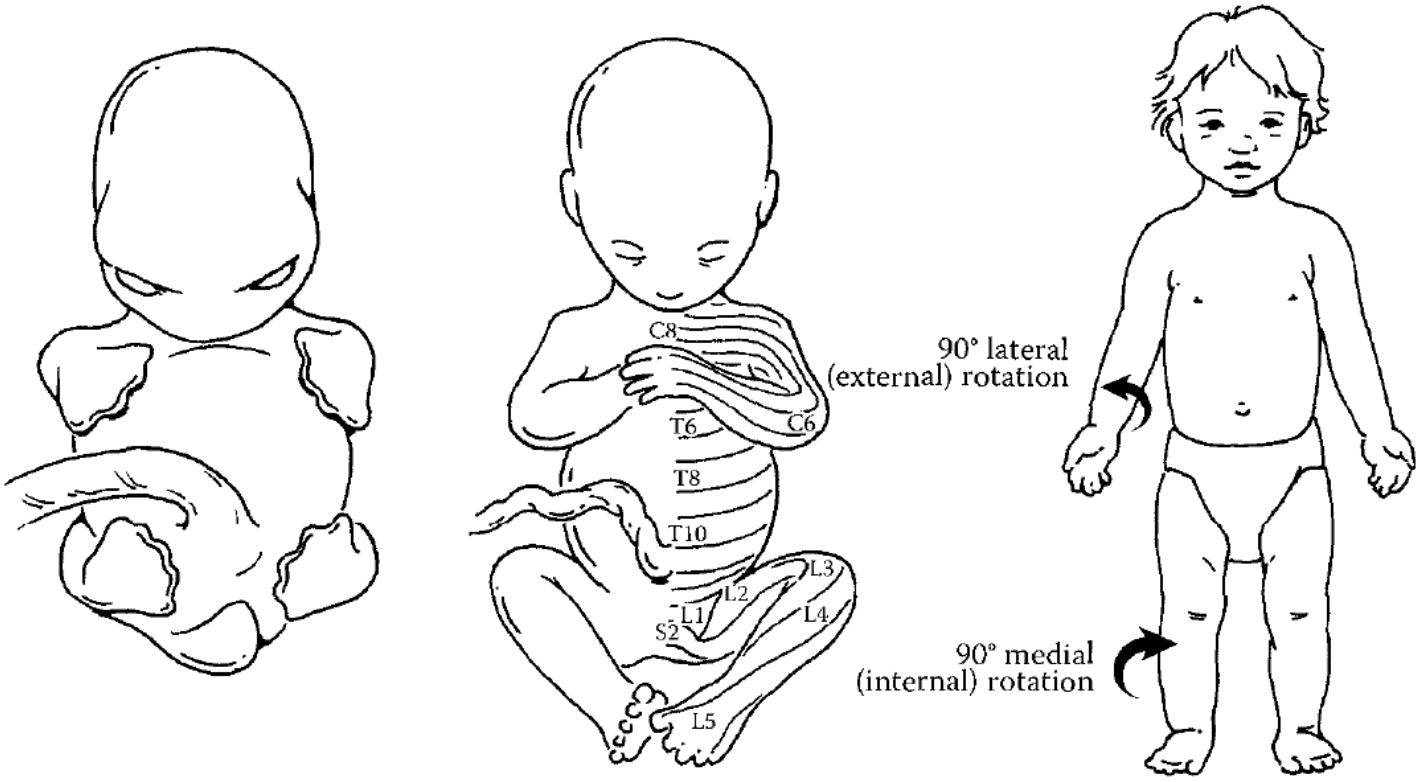




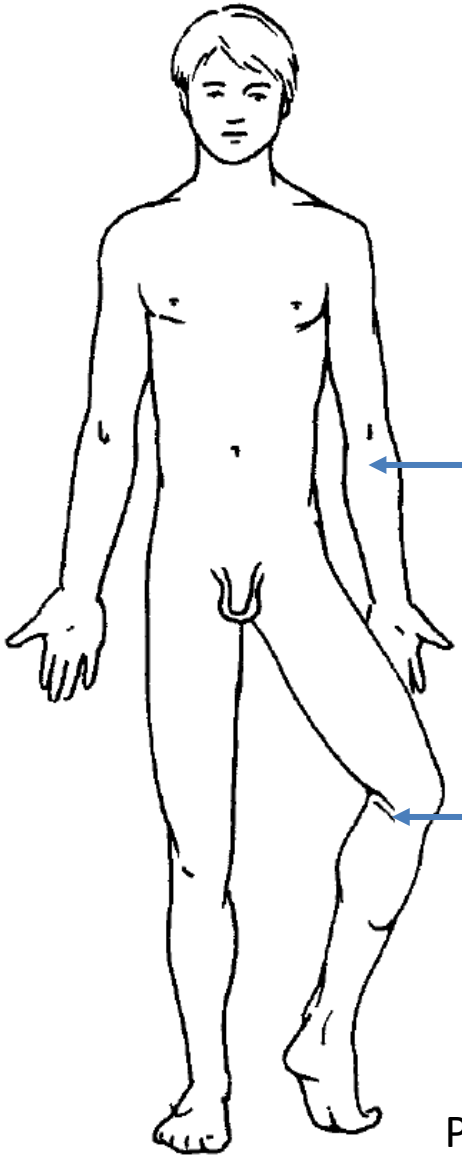
Quadrupedal posture



Two 90 degree rotations occur in development



Lower limb is rotated 180 degrees relative to upper limb



Cubital fossa

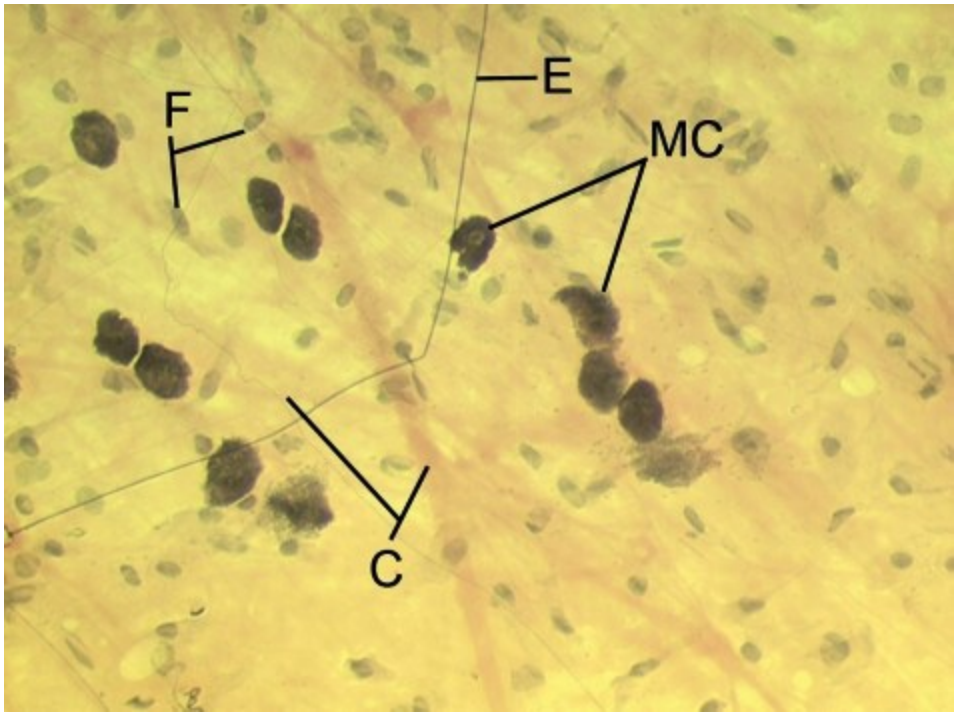
Palmar surface
(Hand)

Popliteal fossa

Plantar surface (Foot)



Superficial Fascia*



C= collagen fibres thick pink
F= fibroblasts
E= elastic fibres thin black
MC= mast cells

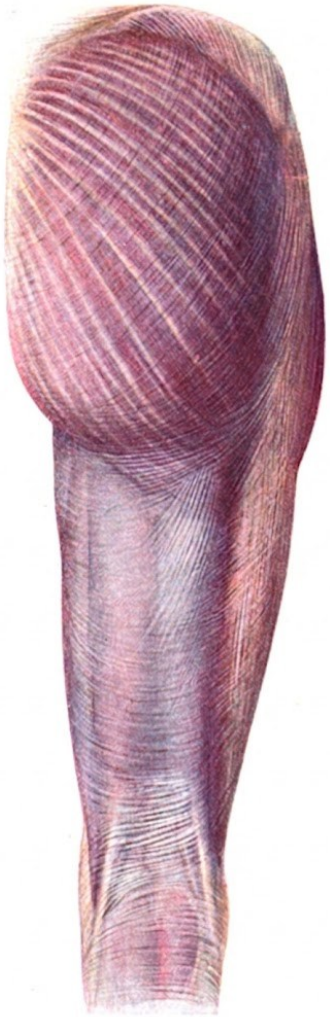
*Fascia = band or bandage (Latin)

FASCIA OF THE LOWER LIMB

Buttocks

Thigh

Leg



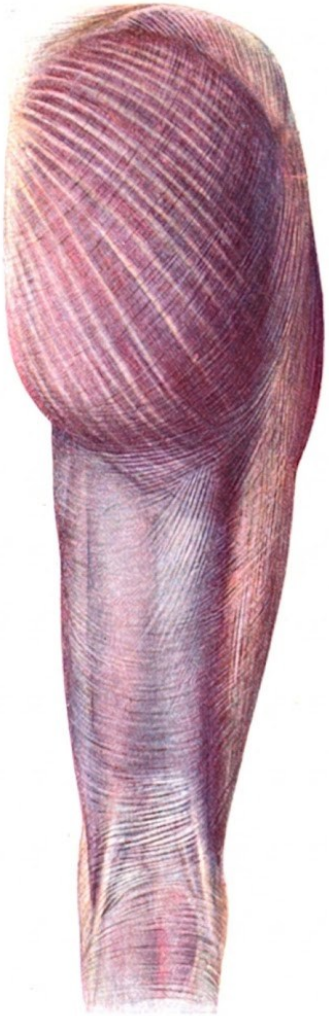
Gluteal fascia

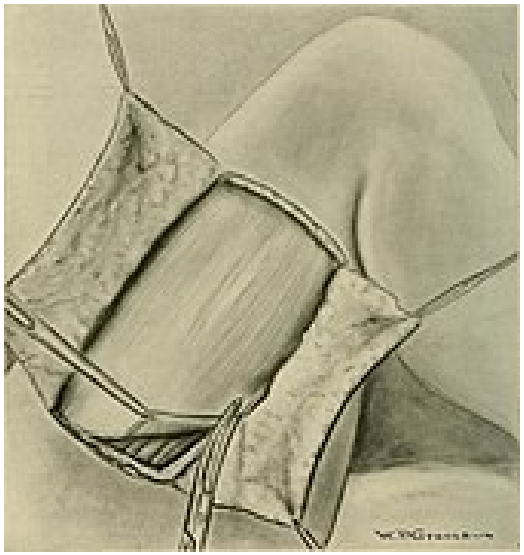
FASCIA OF THE LOWER LIMB

Buttocks

Thigh

Leg





Gluteal fascia

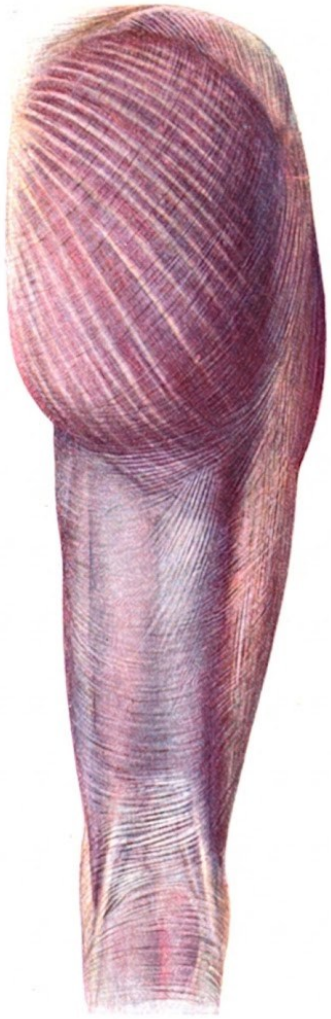
Fascia Lata

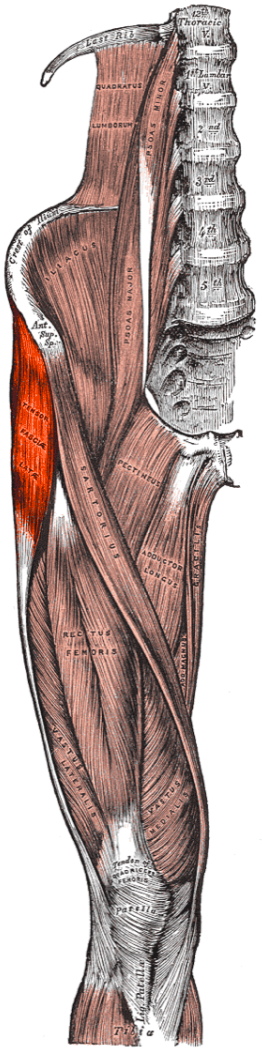
FASCIA OF THE LOWER LIMB

Buttocks

Thigh

Leg



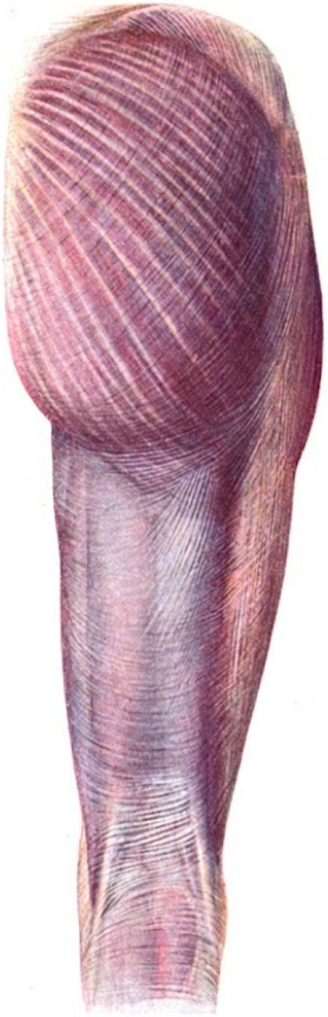


Gluteal fascia

Tensor Fasciae Latae

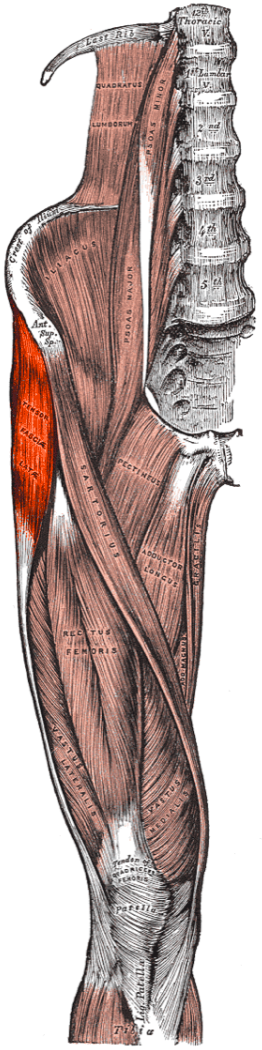
FASCIA OF THE LOWER LIMB

Buttocks



Gluteal fascia

Thigh



Tensor Fasciae Latae

Leg

Proximal

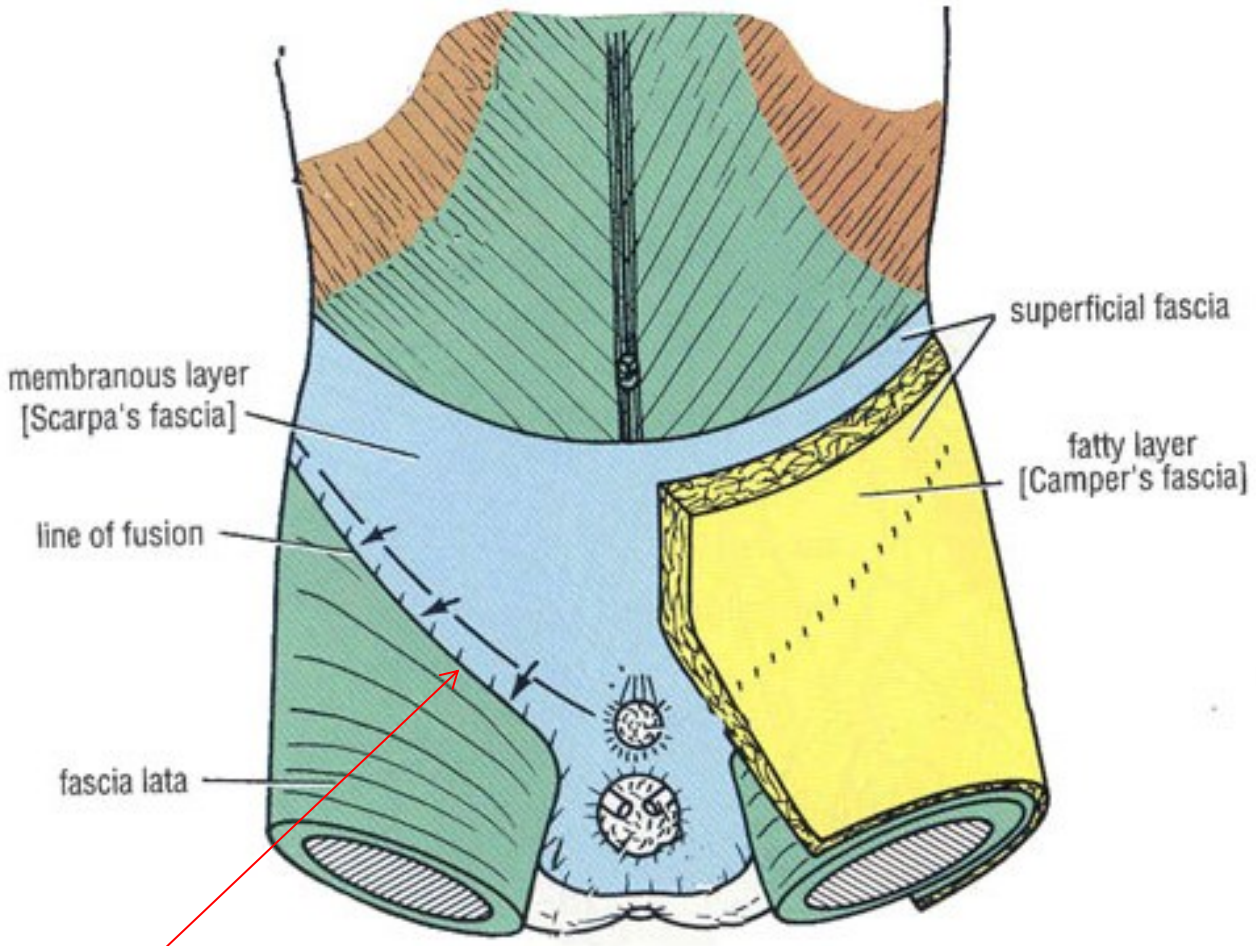


Distal

Crural fascia

white bit

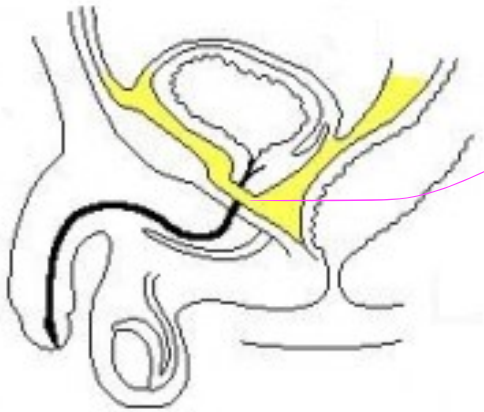
Fascia of abdomen and thigh



Ruptured male urethra

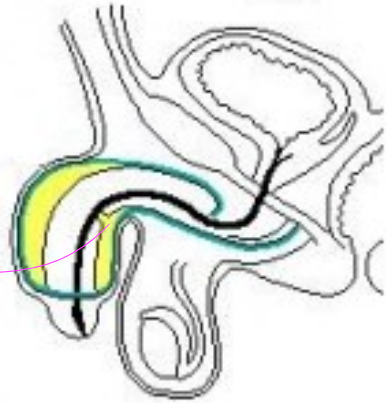
buck's fascia

Intra-pelvic

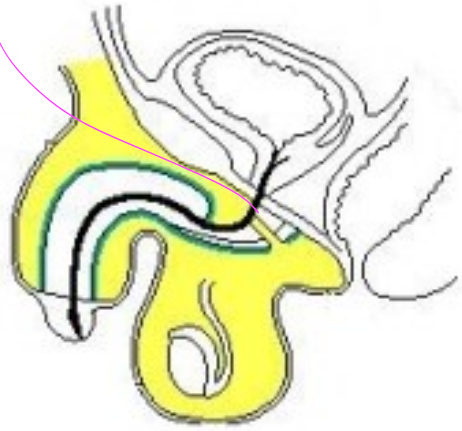


rupture here

Intra-penile

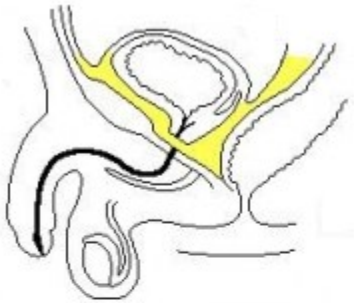


Intra-fascial space

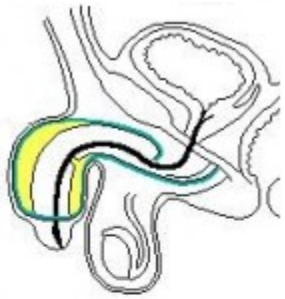


Ruptured male urethra

Intra-pelvic

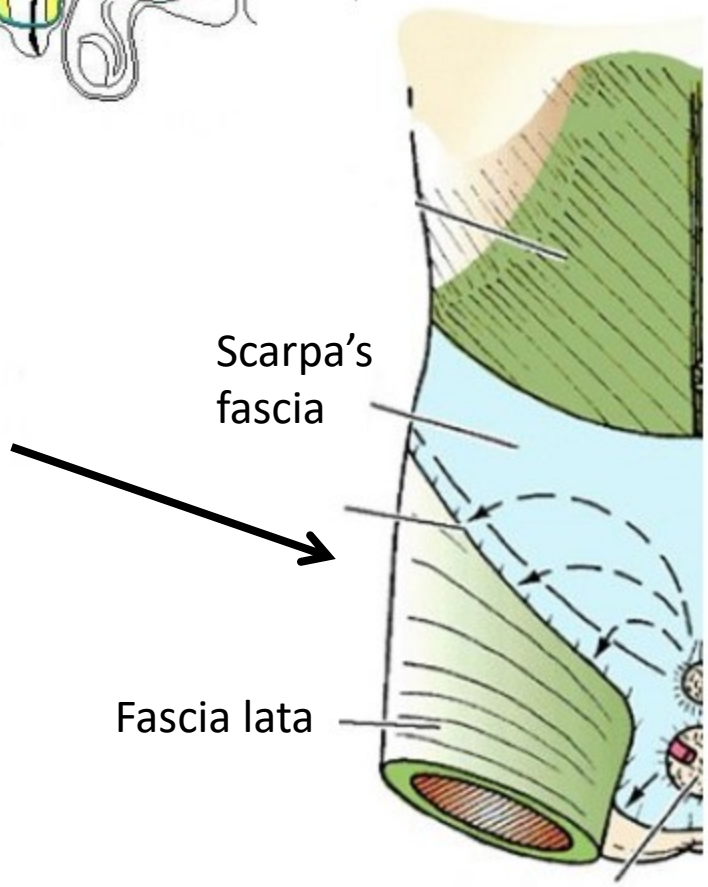


Intra-penile

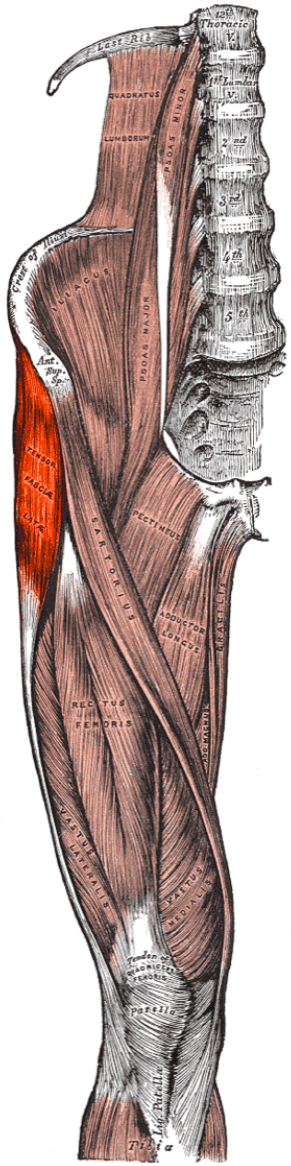


Intra-fascial space

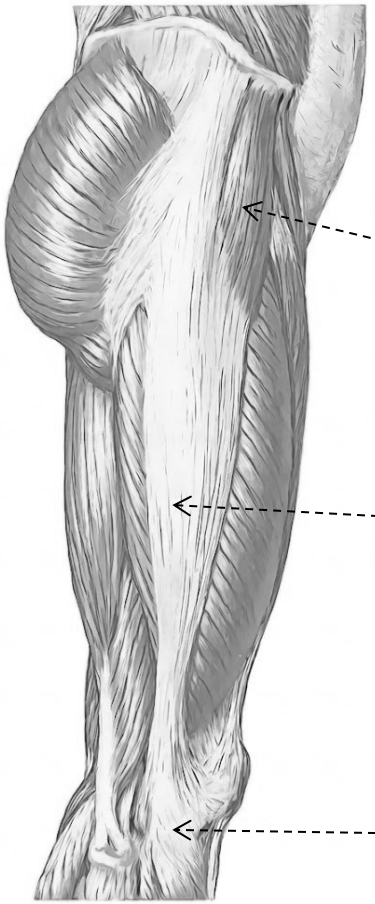




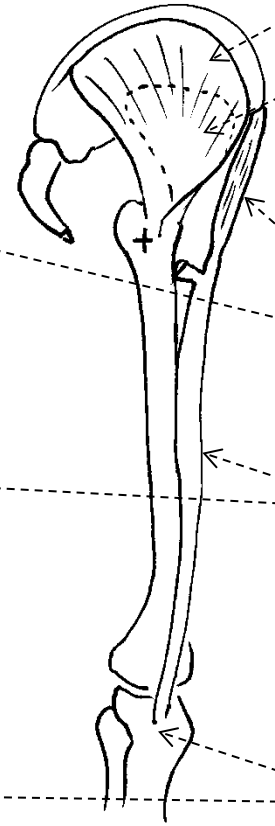
Right Hip Abductors



Anterior view



Right Lateral view



Gluteus medius

Gluteus minimus

Tensor fasciae latae

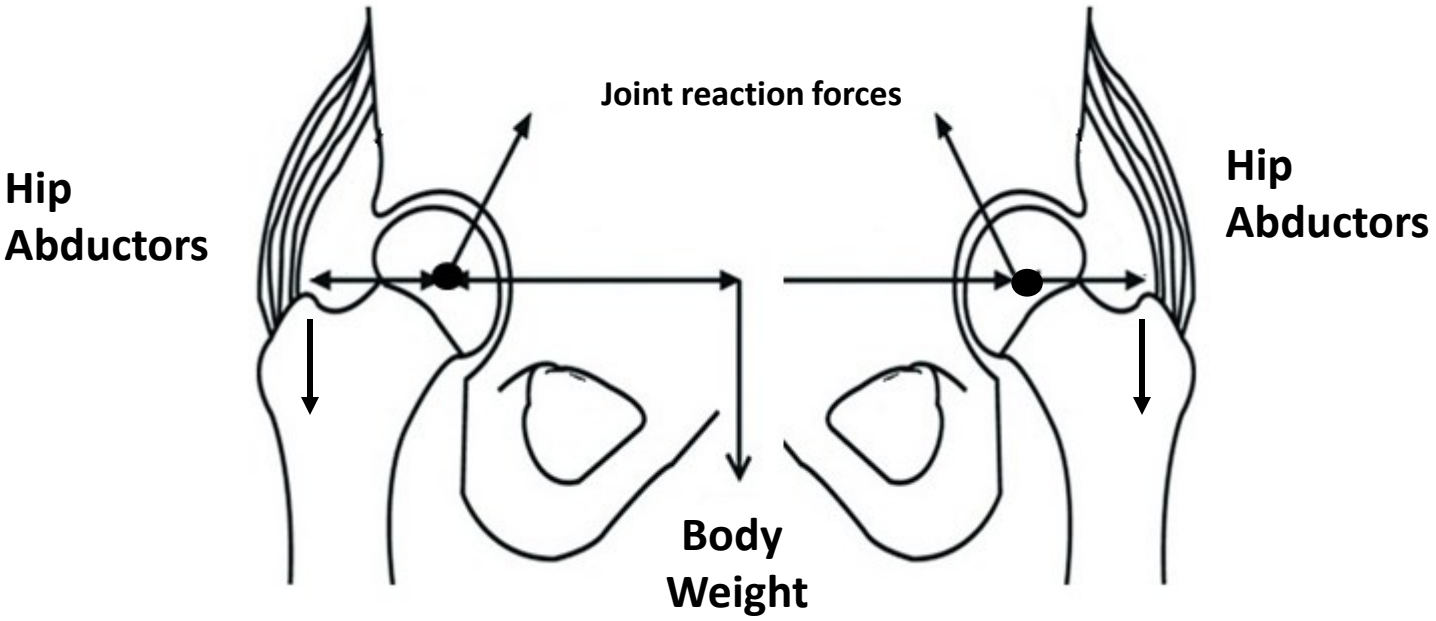
Text

Iliotibial tract

Gerdy's tubercle on tibia

Right Lateral view

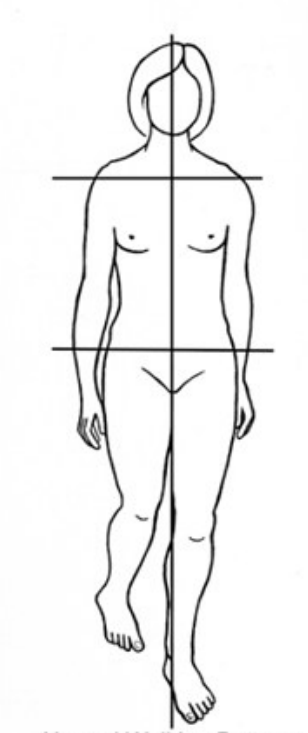
Hip abductors can also stabilize the pelvis



Trendelenburg's Test

Normal





Normal Walking Pattern

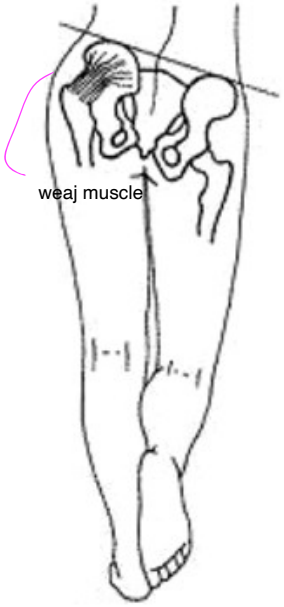
Trendelenburg's Test

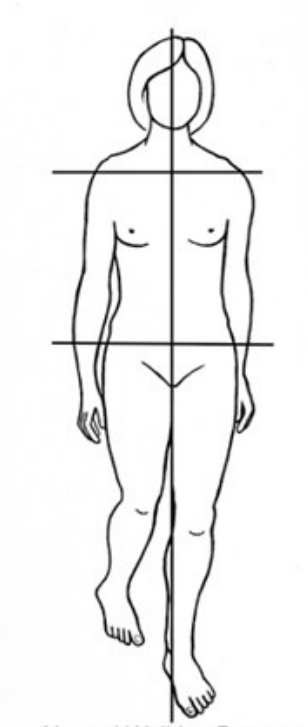
the sound side sags - the strong side sags

Normal



Trendelenburg sign



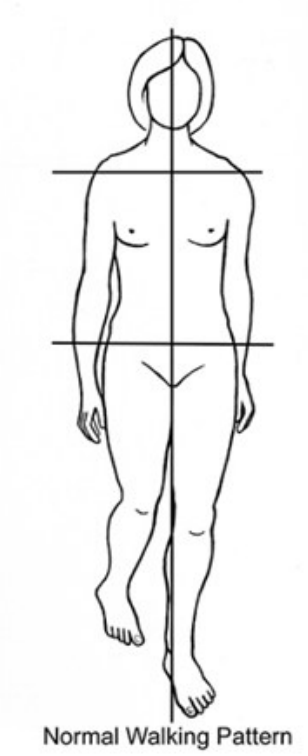


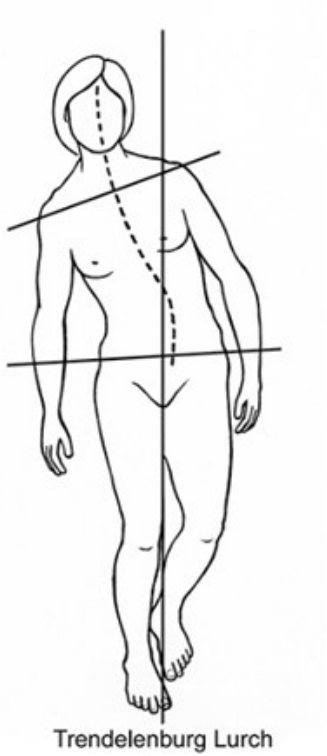
Normal Walking Pattern

Trendelenburg's Test

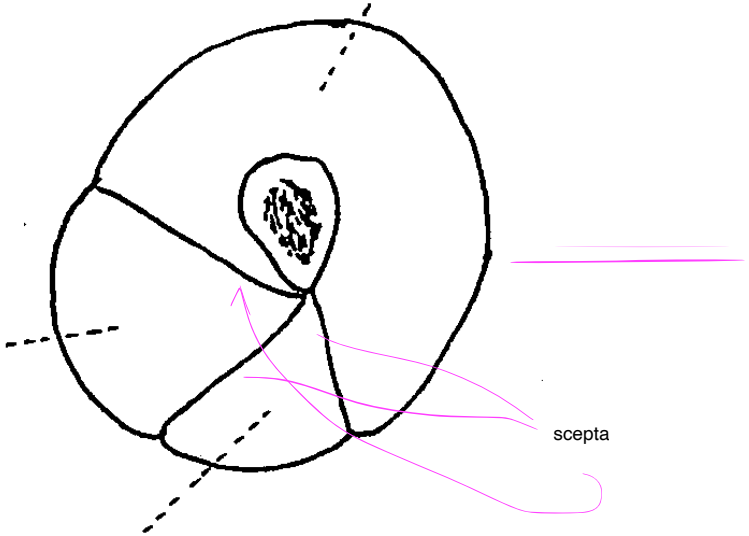





Normal Walking Pattern

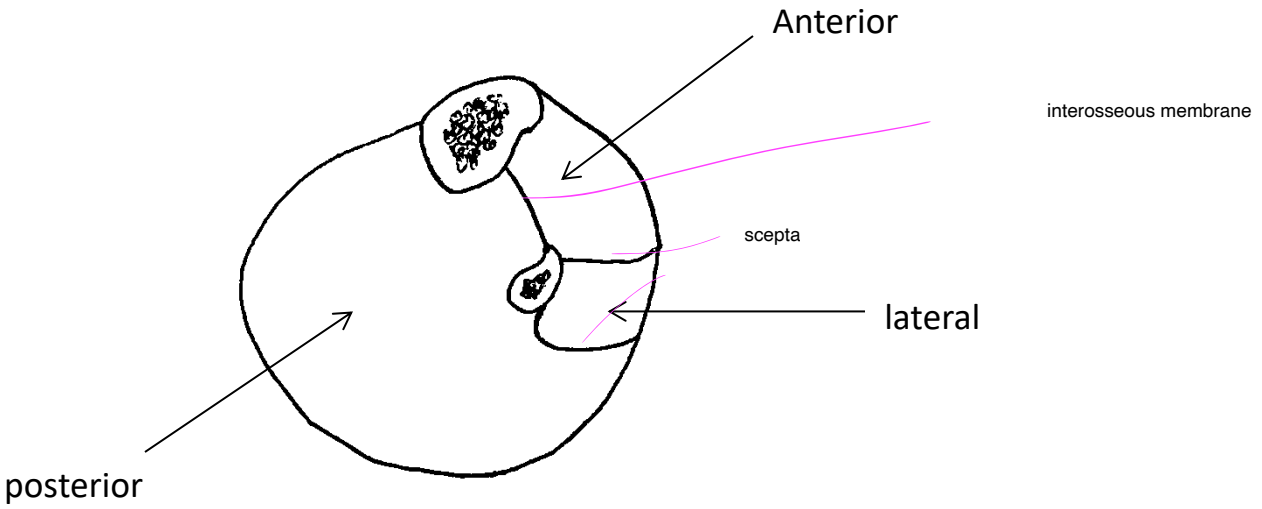

Trendelenburg Lurch

Section through thigh to show fascial compartments



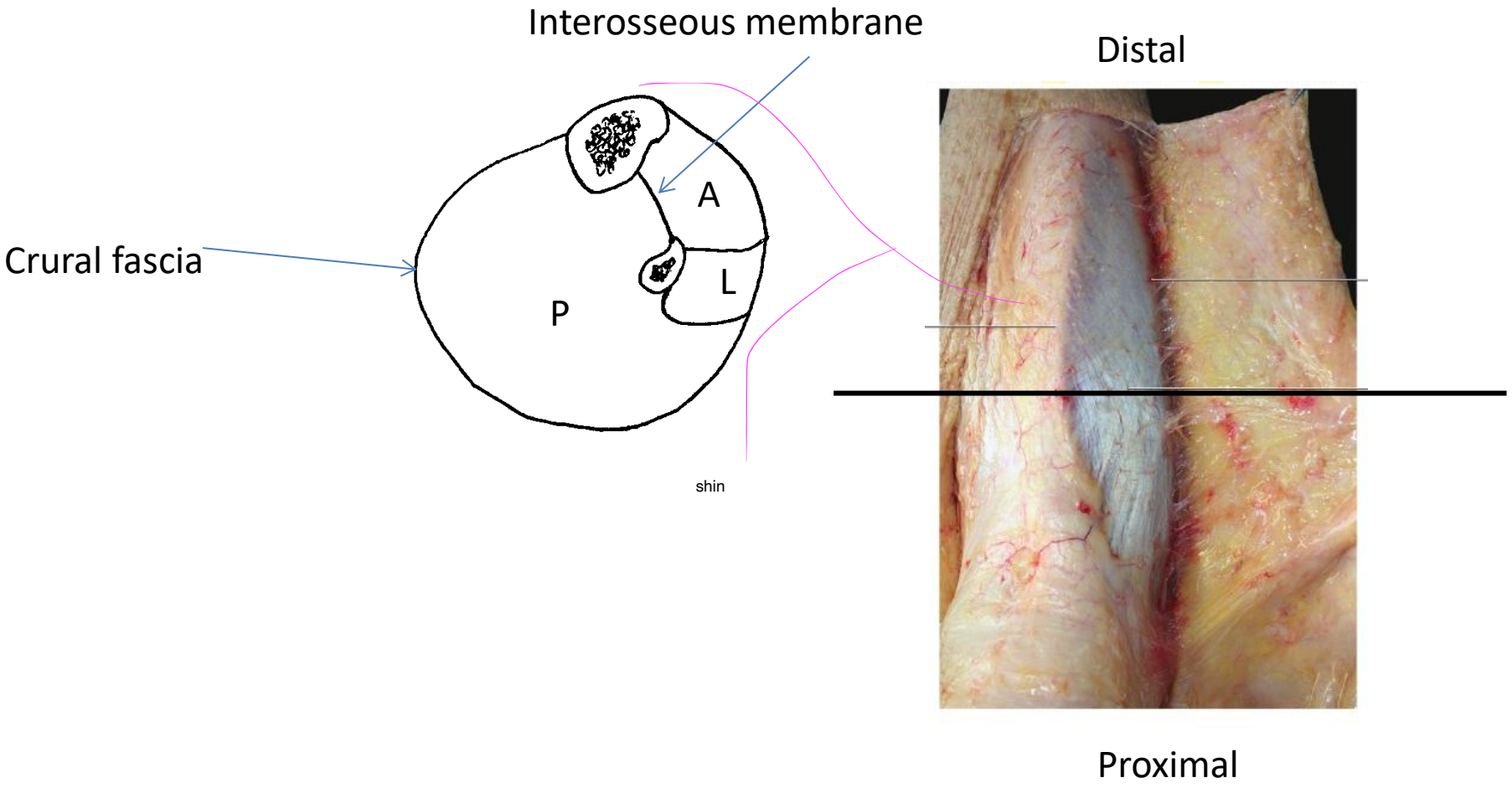
Right Thigh from above

Section through leg to show fascial compartments



Right Leg from above

Compartments of leg



Anterior compartment syndrome



Emergency fasciotomy

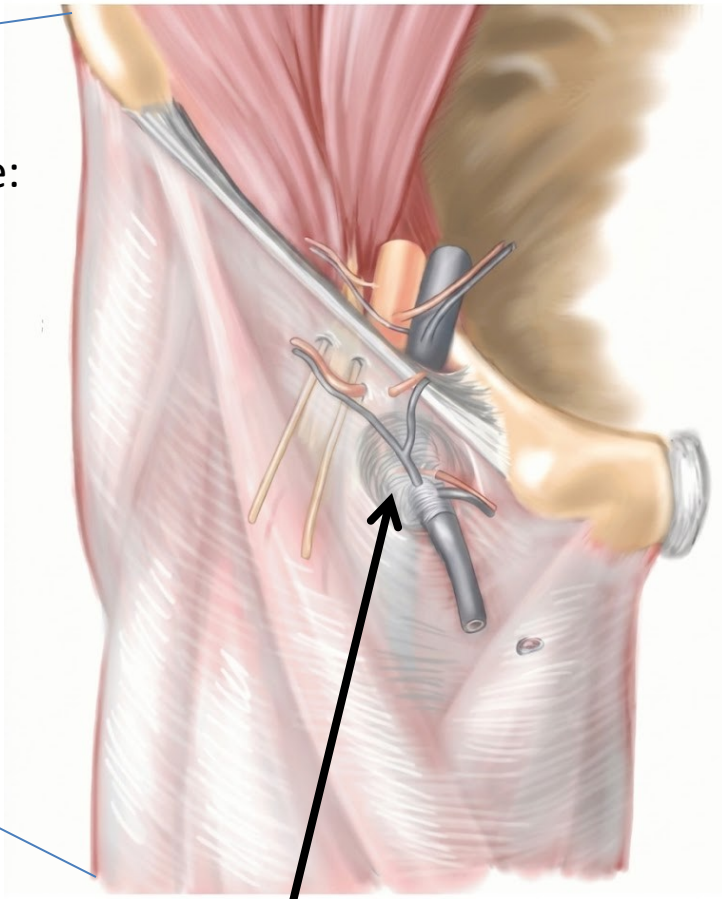


The veins of the lower limb include:

- Superficial (easily seen or “saphenous”) veins

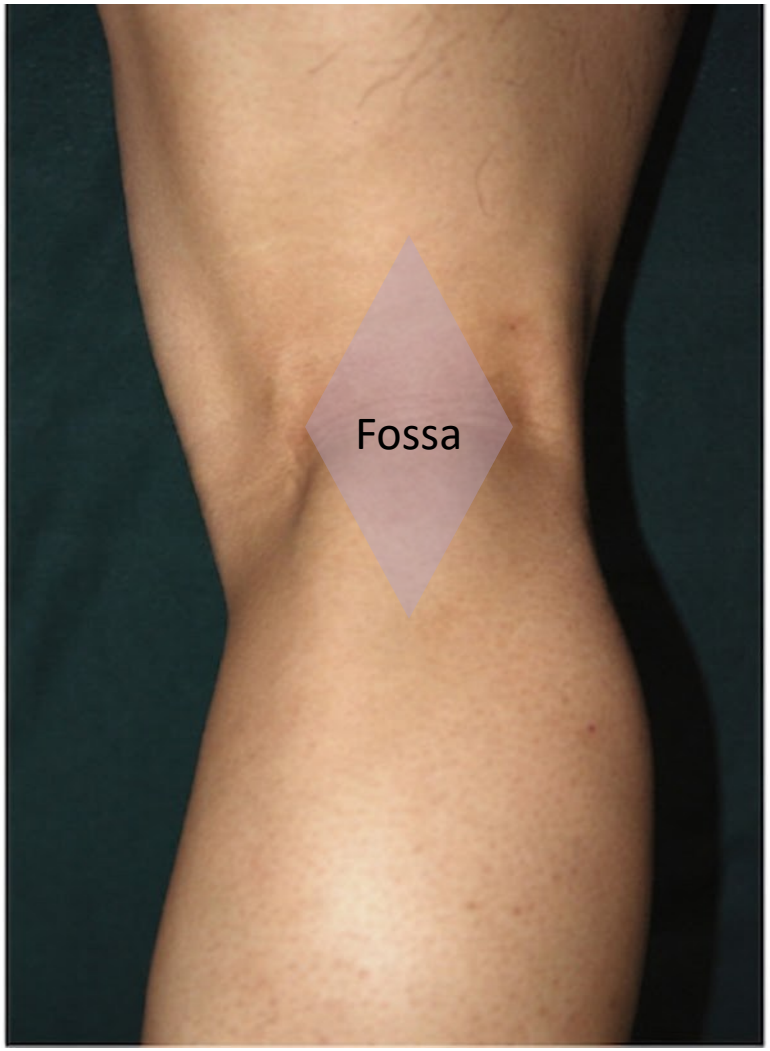
also has an arabic root with a different meaning

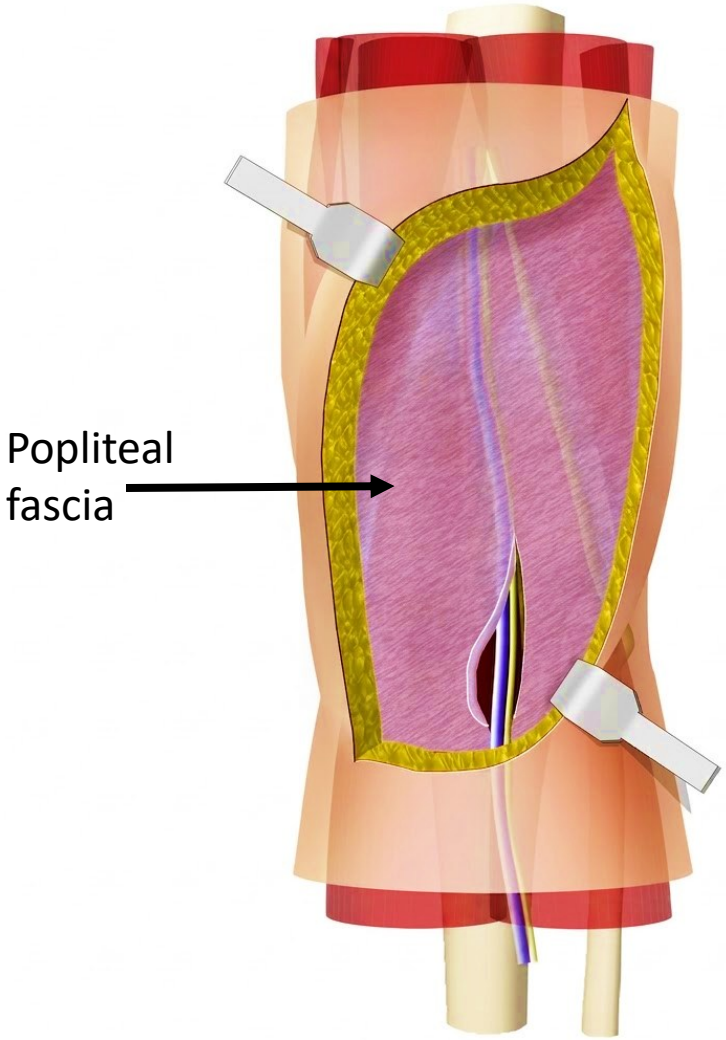
- Deep (not visible) veins



Cribriform (sieve like) fascia
at the saphenous opening facilitates connection between superficial and deep veins

Popliteal fossa





Dissected plantar surface of foot

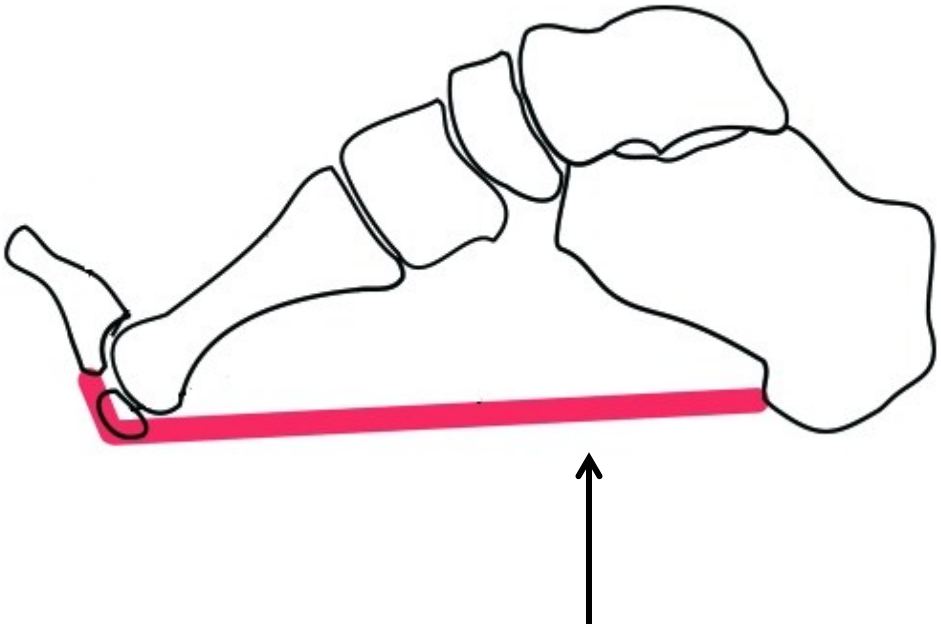
Plantar fascia



WIKIMEDIA COMMONS

Da-wei Chen, Bing Li, Ashwin Aubeeluck, Yun-feng Yang, Yi-gang Huang, Jia-qian Zhou, Guang-rong Yu - Anatomy and Biomechanical Properties of the Plantar Aponeurosis: A Cadaveric Study

Plantar Fascia (a windlass mechanism)



Sample MCQ

Scarpa's membranous layer of the abdomen is anchored at Holden's line to which fascial layer?

A Crural fascia
B Colles's fascia
C Gluteal fascia
D Fascia lata
E Cribriform fascia